

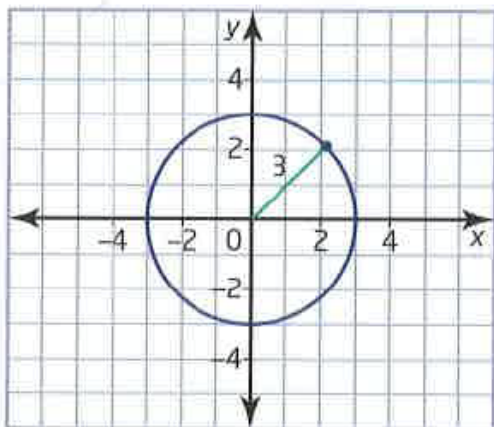
2.4 Equation of a Circle Worksheet

MPM2D

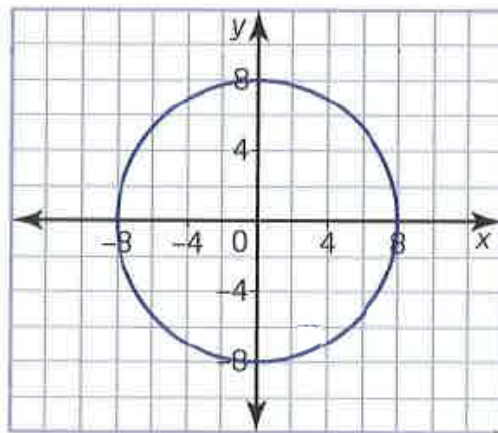
Jensen

1. Determine an equation for each circle

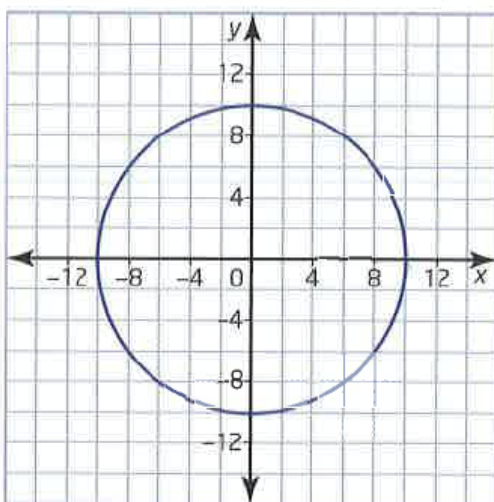
a)



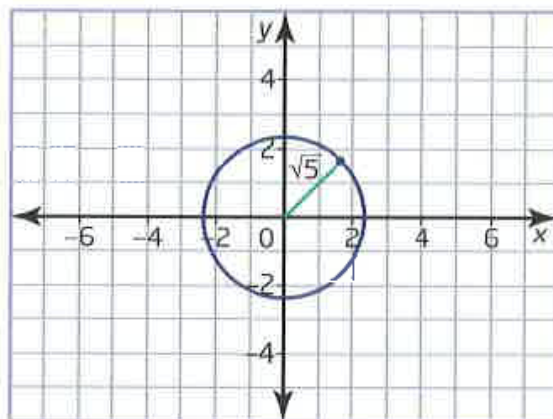
b)

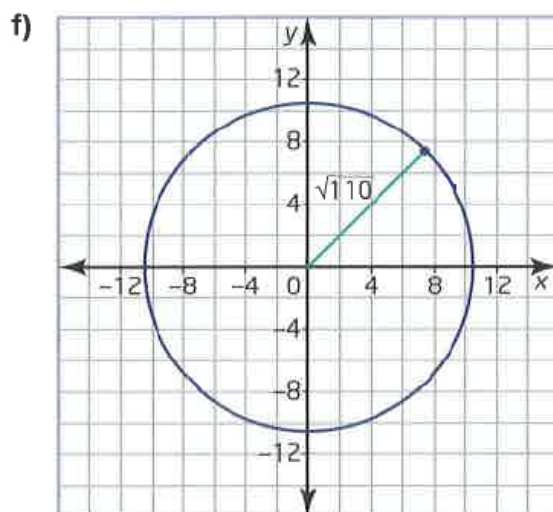
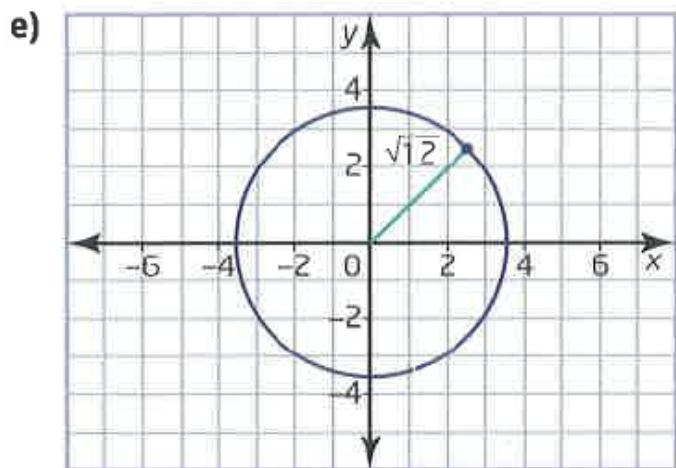


c)



d)





2. For each equation, state the radius of the corresponding circle and give the coordinates of one point on the circle.

a) $x^2 + y^2 = 36$

b) $x^2 + y^2 = 144$

c) $x^2 + y^2 = 20$

d) $x^2 + y^2 = 50$

e) $x^2 + y^2 = 1.69$

3. For each point, find an equation for the circle that is centered at the origin and passes through the point.

a) $(-4, 3)$

b) $(5, 2)$

c) $(-3, -6)$

d) $(-7, 12)$

4. Determine whether each point is on, inside, or outside the circle defined by $x^2 + y^2 = 34$.

a) $(5, -3)$

b) $(4, 4)$

c) $(-6, 0)$

d) $(-3, -5)$

e) $(2, -6)$

f) $(\sqrt{34}, 0)$

5. A farmer is building a circular corral to hold livestock. With distances measured in meters, the shape of the corral is modeled by the equation $x^2 + y^2 = 64$.

a) Find the length of fencing required for this corral.

b) Find the area of the corral.

6. A ship drops its anchor into the water and creates a circular ripple. The radius of this ripple increases at a rate of 50 cm/s.

a) Find an equation for the circle 10 seconds after the anchor is dropped.

b) A small rowboat is 50 meters east and 75 meters north of the point where the anchor was dropped. How long does the ripple take to reach the rowboat?

Answers

1. **a)** $x^2 + y^2 = 9$ **b)** $x^2 + y^2 = 64$
 c) $x^2 + y^2 = 100$ **d)** $x^2 + y^2 = 5$
 e) $x^2 + y^2 = 12$ **f)** $x^2 + y^2 = 110$
2. **a)** 6; points on circle include (6, 0), (0, 6), (-6, 0), and (0, -6)
 b) 12; points on circle include (12, 0), (0, 12), (-12, 0), and (0, -12)
 c) $\sqrt{20}$; points on circle include (2, 4), (-2, 4), (-2, -4), (2, -4), (4, 2)
 d) $\sqrt{50}$; points on circle include (5, 5), (5, -5), (-5, -5), (-5, 5)
 e) 1.3; points on circle include (1.3, 0), (0, 1.3), (-1.3, 0), and (0, -1.3)
3. **a)** $x^2 + y^2 = 25$ **b)** $x^2 + y^2 = 29$
 c) $x^2 + y^2 = 45$ **d)** $x^2 + y^2 = 193$
4. **a)** on **b)** inside **c)** outside
 d) on **e)** outside **f)** on

5. a) 50.3 m b) 201 m²

6. a) $x^2 + y^2 = 250000$ b) 180 seconds